The Long-Term Effects of Indicating AI Fallibility on Automation Bias in Mental Health Application Users

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Figure 1: Seattle Mariners at Spring Training, 2010.

Abstract

CCS Concepts

• **Information systems** → **Users and interactive retrieval**; **Evaluation of retrieval results**; • **General and reference** → **Surveys and overviews**; • **Computing methodologies** → Knowledge representation and reasoning; **Artificial intelligence**; • **Social and professional topics** → Age; Gender; • **Human-centered computing** → Human computer interaction (HCI); **Empirical studies in interaction design**; *Visualization*; **Empirical studies in HCI**.

Keywords

Mental Health, AI, Artificial Intelligence, FWS, STUD, Automation Bias, Fallibility, Mental Health Application, AI Assistant, Long-Term, Long-Term Effects, Priming, Trust, Confidence

ACM Reference Format:
Tim Ruland, Marc Obst, and Nicolas Weber. 2026. The Long-Term Effects of Indicating AI Fallibility on Automation Bias in Mental Health Application Users. In *Proceedings of Abschlusskonferenz FWS/SDUT Sommersemester 2026 (FWS2026).* ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

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1 Introduction

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2 Definitions

2.1 Participants

We define the following sets:

$$\begin{aligned} \text{Groups } G &:= \{\text{Active}, \text{Control}\}, \\ \text{Case Vignettes } C &:= \{\text{Amina}, \text{Aylin}, \text{Lena}, \text{Martina}, \text{Mathias}\}, \\ \text{Truthfulness } T &:= \{\text{Truth}, \text{Untruth}\}. \end{aligned}$$

A *round* is a pair consisting of a case vignettes and a truthfulness label. Hence, the set of all rounds is

$$R := C \times T.$$

A *session* consists of exactly three rounds. Formally, a session s is an ordered triple

$$s = (r_1, r_2, r_3) \in R^3.$$

However, not every element of R^3 is considered valid. A valid session must satisfy the following constraints:

$$|\{i \in \{1,2,3\} \mid r_i = (c, \text{Truth}) \text{ for some } c \in C\}| = 1,$$

i.e. exactly one round in the session is labelled Truth.

Furthermore,

$$\forall i, j \in \{1,2,3\} : \quad i \neq j \Rightarrow c_i \neq c_j,$$

where $r_i = (c_i, t_i)$. That is, no case vignette may occur more than once within the same session.

We denote the set of all valid sessions by

$$S \subseteq R^3.$$

A participant is a triple

$$p = (g, s_1, s_2) \in G \times S \times S,$$

where

$$g \in G, \qquad s_1, s_2 \in S.$$

The two sessions of a participant must share exactly one case vignette. Formally,

$$\left|\left\{c \in C \mid \exists\, t_1, t_2 \in T : (c, t_1) \in \{r_1, r_2, r_3\} \land (c, t_2) \in \{r'_1, r'_2, r'_3\}\right\}\right| = 1,$$

where

$$s_1 = (r_1, r_2, r_3), \qquad s_2 = (r'_1, r'_2, r'_3).$$

Thus:

$$\text{Participants } P := G \times S \times S$$

where both S fulfill all previous conditions

An example of a valid participant is

$$\begin{aligned} p = (&\text{Active}, \\ &((\text{Amina}, \text{Truth}), (\text{Lena}, \text{Untruth}), (\text{Mathias}, \text{Untruth})), \\ &((\text{Martina}, \text{Truth}), (\text{Aylin}, \text{Untruth}), (\text{Mathias}, \text{Untruth}))). \end{aligned}$$

2.2 Participant Allocation

To generate participants satisfying the constraints defined in the previous subsection, we employ a deterministic pseudo-random allocation algorithm, denoted by A_s (participant allocator).

For a fixed pseudo-random seed s, the algorithm defines a mapping

$$A_s : \{0, \ldots, N-1\} \to P$$

where

$$A_s(n) = p_n,$$

with

- P denoting the set of valid participants defined previously,
- $n \in \mathbb{N}_0$ denoting the participant index,
- $p_n \in P$ denoting the generated participants corresponding to index n,
- $s \in \mathbb{N}$ denoting the global pseudo-random seed.

In the present study, the seed value $s = 403$ was used. The algorithm is deterministic with respect to the seed s, thereby ensuring reproducibility of the participant allocation.

The reference Python implementation is provided in the attached file allocate.py. (see A)

2.3 Automation bias

For the purposes of this study, we define a heuristic automation bias score B that serves as an operational measure of automation bias. B is modelled as a continuous function that maps evaluation parameters to a scalar value in the interval $[0, 1]$, representing the degree of evidence that automation bias has occurred.

The function is defined as:

$$B : \{0,1\} \times \{1,2,3,4,5\} \times [0,100] \to [0,1],$$

$$B(i, r, c),$$

where:

- $B(i, r, c) \in [0, 1]$ denotes the automation bias score,
- $i \in \{0, 1\}$ indicates correctness of the automated answer (1 = correct, 0 = incorrect),
- $r \in \{1,2,3,4,5\}$ denotes the evaluator's rating of the answer (1 = very bad, 5 = very good),
- $c \in [0, 100]$ denotes the evaluator's confidence in the rating.

2.3.1 Definition of the Bias Function.

The function is defined as:

$$B(i, r, c) = (1 - i) \cdot \frac{r-1}{5-1} \cdot \frac{c}{100}.$$

2.3.2 Interpretation. The term $(1 - i)$ ensures that automation bias is only considered when the automated answer is incorrect. If the answer is correct ($i = 1$), then $B(i, r, c) = 0$ regardless of the other parameters.

The term $\frac{r-1}{4}$ maps the rating to a normalized interval $[0, 1]$, where higher ratings indicate stronger perceived quality of the answer. In the context of automation bias, higher ratings for incorrect answers indicate a stronger tendency toward bias.

The term $\frac{c}{100}$ represents the evaluator's confidence in their rating. Higher confidence strengthens the evidence of automation bias.

By construction, $B(i, r, c) \in [0, 1]$.

2.3.3 Threshold for Minimal Automation Bias. The threshold value of 0.25 was selected a priori as an operational cutoff corresponding to a neutral evaluation ($r = 3$) combined with moderate confidence ($c = 50$) of an incorrect recommendation. This value was resolved like this:

$$(i, r, c) = (0, 3, 50).$$

Substituting into the function yields:

$$\begin{aligned} B(0, 3, 50) &= (1 - 0) \cdot \frac{3 - 1}{4} \cdot \frac{50}{100} \\ &= 1 \cdot \frac{2}{4} \cdot 0.5 \\ &= 1 \cdot 0.5 \cdot 0.5 \\ &= 0.25. \end{aligned}$$

We therefore define the minimal automation bias threshold as:

$$B_{\text{threshold}} = 0.25, \quad \text{and we consider} \quad B(i, r, c) > 0.25$$

as indicating evidence of automation bias.

3 Methodology

3.1 Study Design

This study employed a mixed longitudinal design consisting of two experimental sessions separated by an interval of one week. Participants were allocated to either the *Active* or *Control* group according to the allocation procedure defined in Section 2.2.

Each participant completed two sessions s_1 and s_2, where each session consisted of three rounds as defined in Section 2.1. In every session, exactly one round contained a truthful AI recommendation and two rounds contained intentionally untruthful recommendations. Furthermore, the two sessions shared exactly one case vignette, as specified by the participant construction constraints.

The primary objective of the study was to investigate the direct and long-term influence of disclaimer messages on participants' evaluations of AI-generated mental-health recommendations and on the occurrence of automation bias.

3.2 Procedure

3.2.1 Session Introduction. At the beginning of the first session, participants provided informed consent and received the following study description:

> "We have prepared an app for AI-supported mental health. It can assess patient cases and provides treatment recommendations."

Participants then selected a nickname that was used to re-identify them during the second session.

Subsequently, demographic information was collected, including age and gender (Male, Female, Non-binary, Other, Prefer not to say).

3.2.2 Pre-Study Questionnaire. Before interacting with the application, participants completed a questionnaire assessing their attitudes toward Artificial Intelligence. The questionnaire contained six items measured on five-point Likert scales:

- Perceived usefulness of AI,
- Acceptability of AI,
- Perceived usefulness of AI in mental-health applications,
- Acceptability of AI in mental-health applications,
- Frequency of AI use in mental-health contexts,
- Frequency of AI use in other contexts.

Higher values indicated greater perceived usefulness, greater acceptability, or more frequent usage, respectively and lower values the opposite, with the middle being medium/unsure.

3.2.3 Round Procedure. For each round, participants were presented with a case vignette describing a mental-health scenario. While the participant reviewed the vignette, the experimenter prepared the application and loaded the corresponding case data.

After reading the vignette, participants indicated that they were ready to proceed and were given access to the application. Participants then submitted a prompt requesting an assessment of the presented case.

The application generated a treatment recommendation based on the underlying case data. Participants reviewed the generated response and subsequently completed a post-interaction questionnaire.

This procedure was repeated for all three rounds of the session.

3.2.4 Post-Round Questionnaire. Following each AI recommendation, participants evaluated the generated answer using the following measures:

- Overall answer quality
  - 1 = Very bad
  - 5 = Very good
- Confidence in the evaluation
  - Scale from 0 to 100
- UEQ+ scales
  - Intuitive Use
  - Trustworthiness of Content
  - Response Behaviour
  - Response Quality
  - Clarity
  - Risk Handling

3.3 Experimental Conditions

3.4 Session 1

Participants assigned to the *Active* group received two disclaimer interventions during Session 1:

(1) A full-screen disclaimer before interacting with the application.
(2) An inline disclaimer is displayed while waiting for the AI-generated response.

Participants assigned to the *Control* group did not receive either disclaimer.

3.5 Session 2

One week after completing Session 1, participants returned for Session 2.

Participants again provided consent, were reminded of the study context, and were re-identified using their previously selected nickname. They then completed the same AI-attitude questionnaire and proceeded through the three rounds assigned to their second session.

To investigate potential persistence effects, no disclaimer messages were shown during Session 2, regardless of the participant's original group assignment. All participants therefore interacted with the application under identical conditions during the second session.

3.6 Automation Bias Assessment

Automation bias was operationalized using the bias score $B(i, r, c)$ defined in Section 2.3. For each evaluated AI recommendation, the correctness of the recommendation (i), the participant's quality rating (r), and the participant's confidence (c) were recorded.

Bias scores exceeding the predefined threshold

$$B(i, r, c) > 0.25$$

were considered indicative of automation bias.

4 Hypotheses

We conducted our study under the following hypotheses:

- H_1 Exposure to a design intervention in a mental health AI application will lead to a sustained reduction in automation bias, such that participants in the intervention group also will exhibit lower automation bias scores than participants in the control group during a subsequent usage session in which no additional intervention is provided.
- H_0 There will be no differences between the intervention and control groups in terms of changes in automation bias over time. The design intervention will not lead to a long-term effect on automation bias.

5 Results

Acknowledgments

We would like to thank our instructor, Rosbach, E., for their guidance and support throughout this project.
We also thank all testers who participated in the study and made this work possible.

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References

[1] [SW] The CGAL Project, The Computational Geometry Algorithms Library (Coord.by CGAL Editorial Board), 1996. URL: https://cgal.org/.

A Participant Allocation Python Implementation

allocate.py